

Experiment 15: Iris Flower Classification using Naive Bayes

Aim

To implement **Iris Flower Classification** using the **Naive Bayes classifier**.

Python Program

```
import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

import seaborn as sns

import matplotlib.pyplot as plt

iris = load_iris()

X = pd.DataFrame(iris.data, columns=iris.feature_names)

y = iris.target

print("Dataset:")

print(X.head())

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:", accuracy * 100, "%")
print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Iris Classification")
plt.show()
new_flower = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_flower)
print("\nNew Flower Prediction:", iris.target_names[prediction[0]])
```

Sample Output

Dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Accuracy: 100.0 %

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Confusion Matrix:

```
[[10 0 0]
 [ 0 9 0]
 [ 0 0 11]]
```

